

Standard Form & Indices

Test 8 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 6^{11} (1)

A2. 2^6 (1)

A3. 9^{12} (1)

A4. $3^3 \times 3^1 = 3^4$ (1) = **81** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 8^2 = 1 \div 64$ (1) = **1/64** (1)

B3. $1 \div 5^1 = 1 \div 5$ (1) = **0.2** (1)

Section C — Writing standard form

C1. 6.3 (1) = **6.3×10^4** (1)

C2. 1.7 (1) = **1.7×10^6** (1)

C3. 2.9 (1) = **2.9×10^{-3}** (1)

C4. 4 (1) = **4×10^{-6}** (1)

Section D — Converting back, and calculating

D1. **260,000** (1)

D2. **0.0008** (1)

D3. $6 \times 7 = 42$, $10^5 \times 10^2 = 10^7$ (1) = 42×10^7 (1) adjust to standard form: **4.2×10^8** (1)

D4. $2 \div 8 = 0.25$, $10^9 \div 10^3 = 10^6$ (1) = 0.25×10^6 (1) adjust to standard form: **2.5×10^5** (1)

Section E — Powers to know, and estimating roots

E1. **64** (1)

E2. $4^2=16$, $5^2=25$ = between **4 and 5** (1)

Section F — Laws of indices with algebra

F1. **g^{14}** (1)

F2. **h^9** (1)

F3. **a^{14}** (1)

F4. $7 \times 3 = 21$ (1) $b^{3+5} = b^8$ (1) = **$21b^8$**

F5. $3^2 = 9$ (1) $c^{4 \times 2} = c^8$ (1) = **$9c^8$**

Section G — Fractional indices (Higher)

G1. $\sqrt{121} = \mathbf{11}$ (1)

G2. cube root of 512 = **8** (1)

G3. $\sqrt{4} = 2$ (1) $2^5 = \mathbf{32}$ (1)

G4. cube root of 27 = 3 (1) $1 \div 3 = \mathbf{1/3}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.